

Word Equations with Length Constraints: What's Decidable?

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Abstract. We prove several decidability and undecidability results for the satisfiability and validity problems for languages that can express solutions to word equations with length constraints. The atomic formulas over this language are equality over string terms (word equations), linear inequality over the length function (length constraints), and membership in regular sets. These questions are important in logic, program analysis, and formal verification. Variants of these questions have been studied for many decades by mathematicians. More recently, practical satisfiability procedures (aka SMT solvers) for these formulas have become increasingly important in the context of security analysis for string-manipulating programs such as web applications.

We prove three main theorems. First, we give a new proof of undecidability for the validity problem for the set of sentences written as a $\forall\exists$ quantifier alternation applied to positive word equations. A corollary of this undecidability result is that this set is undecidable even with sentences with at most two occurrences of a string variable. Second, we consider Boolean combinations of quantifier-free formulas constructed out of word equations and length constraints. We show that if word equations can be converted to a *solved form*, a form relevant in practice, then the satisfiability problem for Boolean combinations of word equations and length constraints is decidable. Third, we show that the satisfiability problem for quantifier-free formulas over word equations in *regular solved form*, length constraints, and the membership predicate over regular expressions is also decidable.

1 Introduction

The complexity of the satisfiability problem for formulas over finite-length strings (theories of strings) has long been studied, including by Quine [23], Post, Markov and Matiyasevich [17], Makanin [15], and Plandowski [12, 20, 21]. While much progress has been made, many questions remain open especially when the language is enriched with new predicates.

Formulas over strings have become important in the context of automated bugfinding [8, 25], and analysis of database/web applications [7, 14, 27]. These program analysis and bugfinding tools read string-manipulation programs and generate formulas expressing their results. These formulas contain equations over string constants and variables, membership queries over regular expressions, and inequalities between string lengths.

In practice, formulas of this form have been solved by off-the-shelf satisfiability procedures such as HAMPI [8, 13] or Kaluza [25]. In this context, a deeper understanding of the theoretical aspects of the satisfiability problem for this class of formulas may be useful in practice.

Problem Statement: We address three problems. First, what is a boundary for decidability for fragments of the theory of word equations? Namely, is the $\forall\exists$ -fragment of the theory of word equations decidable? Second, is the satisfiability problem for quantifier-free formulas over word equations and the length function decidable under some minimal practical conditions? Third, is the satisfiability problem for quantifier-free formulas over word equations, the length function, and regular expressions decidable under some minimal practical conditions?

The question of whether the satisfiability problem for the quantifier-free theory of word equations and length constraints is decidable has remained open for several decades. Our decidability results are a partial and conditional solution. Matiyasevich [18] observed the relevance of this question to a novel resolution of Hilbert's Tenth Problem. In particular, he showed that if the satisfiability problem for the quantifier-free theory of word equations and length constraints is undecidable, then it gives us a new way to prove Matiyasevich's Theorem (which resolved the famous problem) [17, 18].

Summary of Contributions

1. We show that the validity problem (decision problem) for the set of sentences written as a $\forall\exists$ quantifier alternation applied to positive word equations (i.e., AND-OR combination of word equations without any negation) is undecidable. (Section 3)
2. We show that if word equations can be converted to a *solved form* then the satisfiability problem for Boolean combinations of word equations and length constraints is decidable. (Section 4)
3. The above-mentioned decidability result has immediate practical impact for applications such as bug-finding in JavaScript and PHP programs. We empirically studied the word equations in the formulas generated by the Kudzu JavaScript bugfinding tool [25] and verified that most word equations in such formulas are either already in solved form or can be automatically and easily converted into one. (Section 4)
4. We further show that the satisfiability problem for quantifier-free formulas constructed out of Boolean combinations of word equations in *regular solved form* with length constraints and the membership predicate for regular sets is also decidable. This is the first such decidability result for this set of formulas. (Section 5)

We now outline the layout of the rest of the paper. In Section 2 we define a theory of word equations, length constraints, and regular expressions. In Section 3 we prove the undecidability of the theory of $\forall\exists$ sentences over positive word equations. In Section 4 (resp. Section 5) we give a conditional decidability result for the satisfiability problem for the quantifier-free theory of word equations and length constraints (resp. word equations, length constraints, and regular expressions). Finally, in Section 6 we provide a comprehensive overview of the decidability/undecidability results for theories of strings over the last several decades.